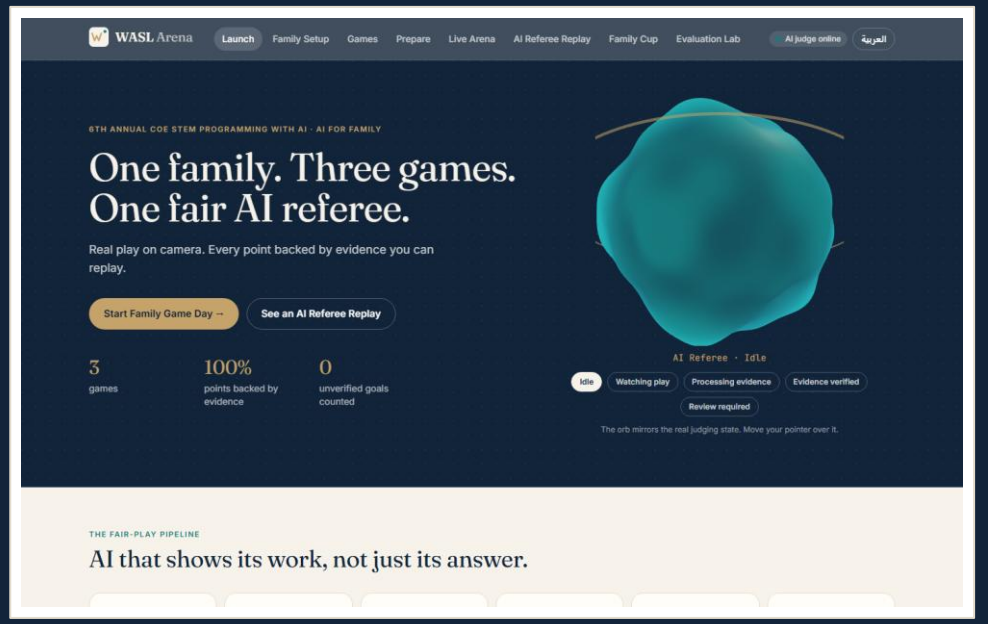
6TH ANNUAL COE STEM PROGRAMMING WITH AI · CYCLE 3 · AI FOR FAMILY

WASL Arena

One family. Three games. One fair AI referee.



Family AI Game Day · Football AI · Falaj Flow · Souq Signal
Team 204 · Team code AYUCK · Track: Cycle 3 (grades 9-12)
May Ahmed Mourad (G12) · Shamsa Faris Almazrouei (G12) · Fatima Mohamed Almazrouei (G11)

VISION

A fair referee for every family game

- Real physical play: ball, board, cards, voices
- An AI that **watches and listens**, then applies written rules
- Every point **linked to replayable evidence**
- A human host always has the final word
- Culture built in: **falaj** irrigation and the **souq** market

Fair play = evidence + doubt

If the AI cannot see it, it does not score it. Uncertain calls wait for the host.

PROBLEM

Family games end in arguments

“Was that a goal?”

Nobody agrees what the camera saw.

“You said it after!”

Speech and action are hard to line up by memory.

“Is the score right?”

Manual tallies drift; nobody can check them.

- Black-box AI scoring fails the trust test
- An AI that guesses is worse than no AI
- Families need **explanations**, not just answers

CONTEXT

Why family gaming matters

- Shared play builds **connection across generations**
- Physical games keep families **moving and talking**
- Heritage games pass on **culture** (falaj, souq)
- Fair, simple judging lets **children and adults compete together**
- Screens act as **host and referee**, not the entertainment

Our principle

Technology should make the family game more fun and more fair, never take it over.

The official challenge

- Working website/app for family games inspired by **culture, heritage or sports**
- AI observes real play through **video and audio**, applies rules, keeps score
- **Cycle 3:** football + 2 original games (team and individual)
- Football: track players and ball, goals, **possession, MVP**
- Upload video **or** record by camera + microphone
- Human **host** sets up, runs rounds/turns/timers, shows reasons

Excluded examples

Heritage guessing · cup stacking · ring toss · copy-the-rhythm. Our games are original.

SOLUTION

Our solution: WASL Arena

Football AI

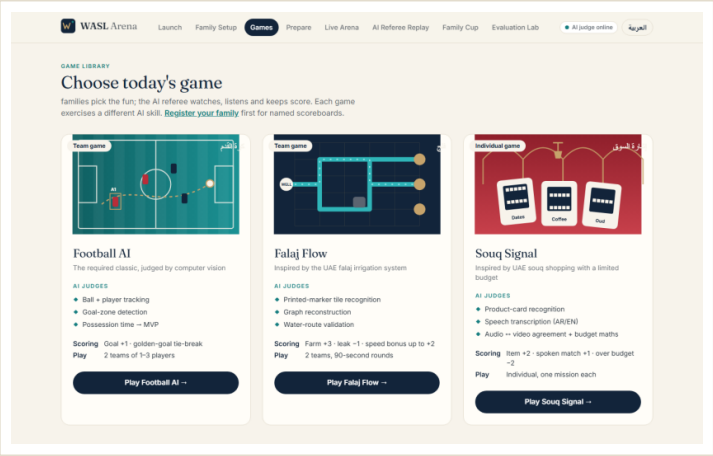
Team · tracking, goals, possession, MVP

Falaj Flow

Team · board vision + graph reasoning

Souq Signal

Individual · vision + speech + budget



- 8 screens, bilingual EN/AR
- Real video/audio judging
- Host review + audit trail

WHY IT WINS

Value proposition

Families

Fair, fast scoring that everyone can replay and understand.

Hosts

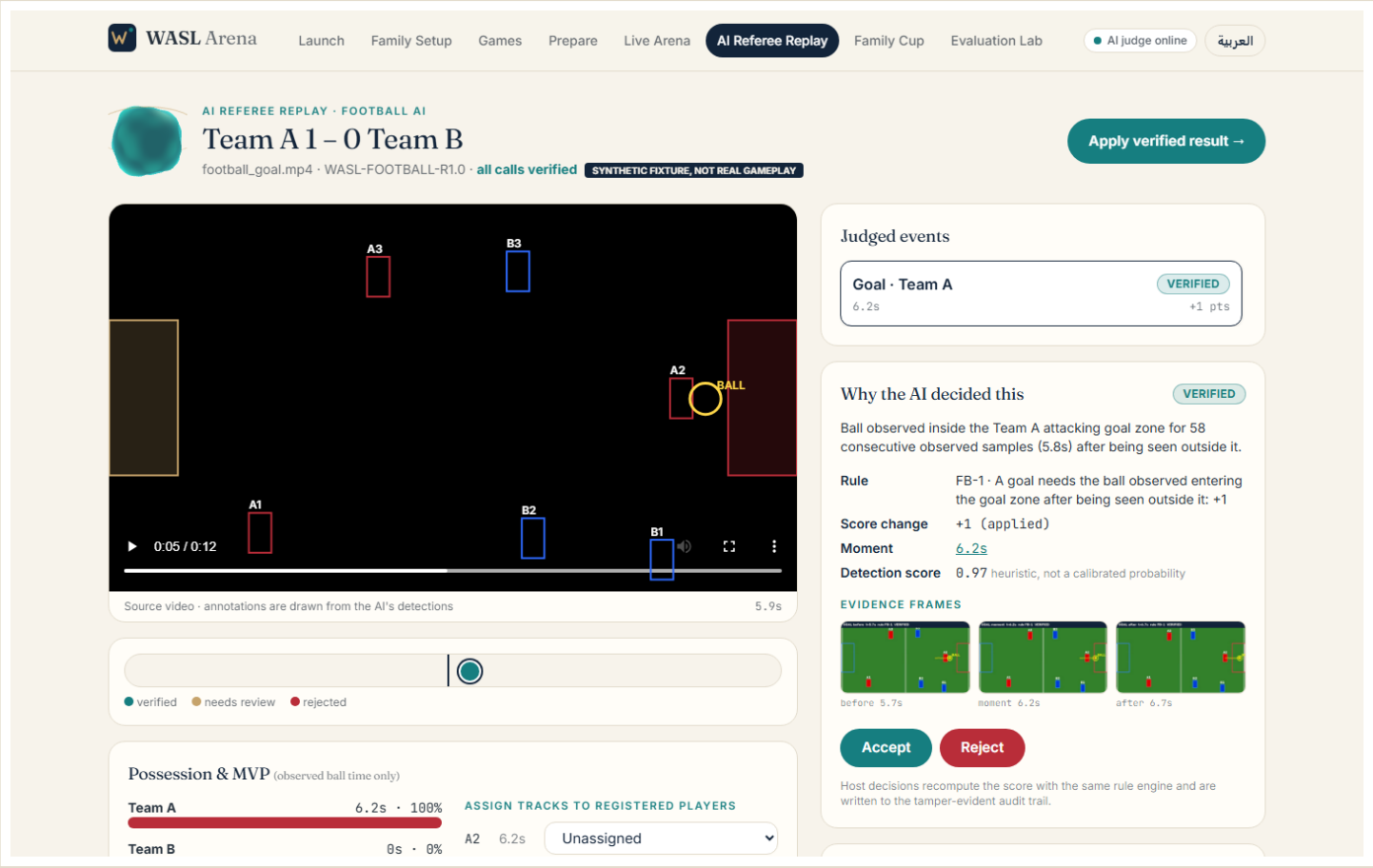
Setup, rounds, timers, review and results in one console.

Judges

Real pipelines, tests and honest limits you can inspect.

- Explainable: rule, frames, timestamps, score change
- Honest: NEEDS_REVIEW instead of guessing
- Tamper-evident: hashes + signatures + audit chain

Signature innovation: AI Referee Replay



- **Source video** with live annotations
- **Timestamps** and evidence frames
- **Rule applied** and score change
- **Verification status** + uncertainty
- **Human review** accept / reject

HOST FLOW

User journey

1 Choose game

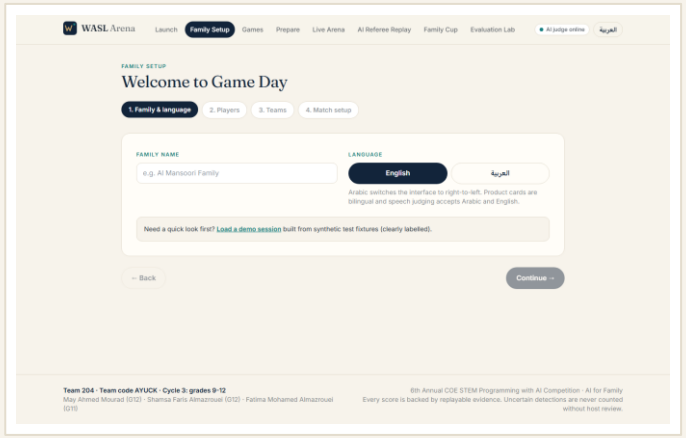
2 Players & teams

3 Rules & ready

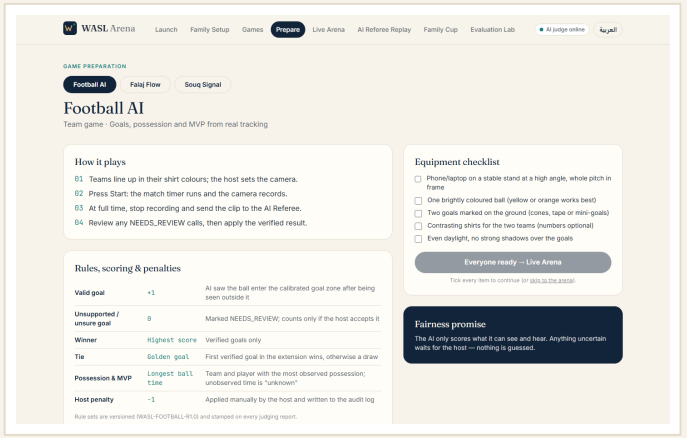
4 Round in play

5 AI judges

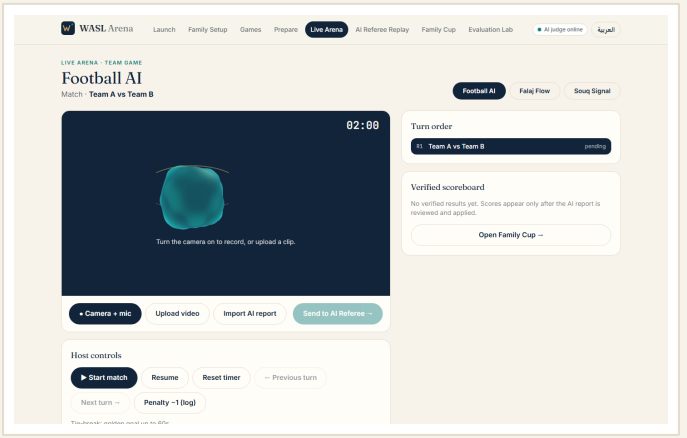
6 Final results



Family setup



Preparation



Live arena

The three-game collection

Football AI · Team

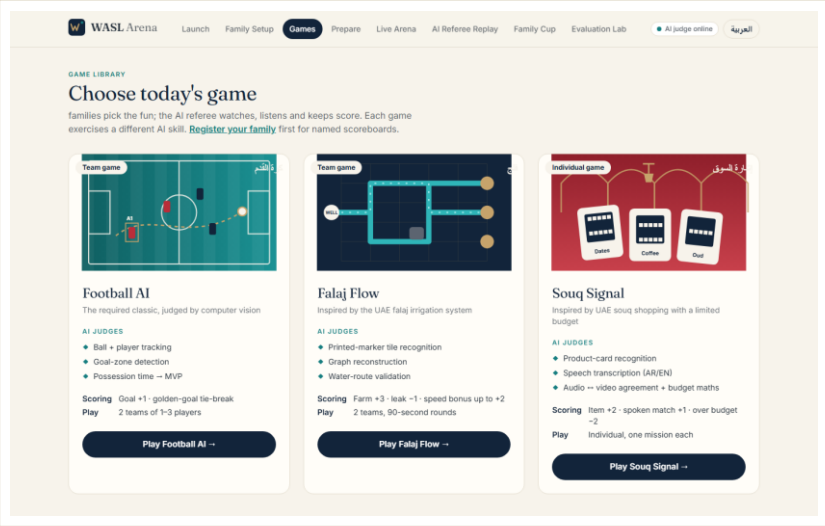
Tracking, goal zones, possession, MVP

Falaj Flow · Team

ArUco tile reading, graph, water routes

Souq Signal · Individual

Cards + speech + budget agreement

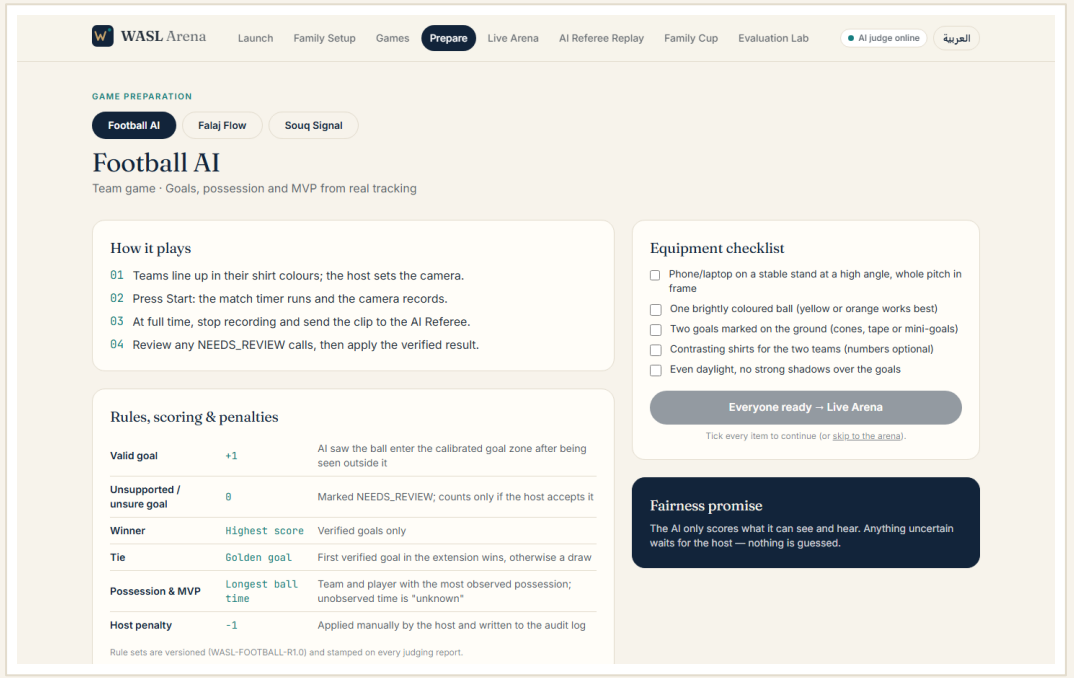


- Team **and** individual play
- No excluded example games
- Printable boards and cards included

GAME 1 · TEAM

Football AI: overview

- Mini-match, fixed camera, bright ball, marked goals
- Valid goal **+1**; unsupported goal **0**
- Winner: highest verified score
- Tie: configurable **golden goal**
- **MVP** = longest verified possession
- Host penalty −1 (audit-logged)



Goal-zone calibration in Game Preparation

HOW IT SEES

Football AI: tracking

- Frames sampled at **10 fps**
- Ball = best **circular contour** in a calibrated colour range
- Players = **team-colour blobs**, tracked frame to frame
- Only **observed** positions count: no interpolation
- Players are tracks; the host maps tracks to names



Tracked ball trail and player boxes (synthetic fixture)

EVIDENCE

Football AI: referee replay



The moment



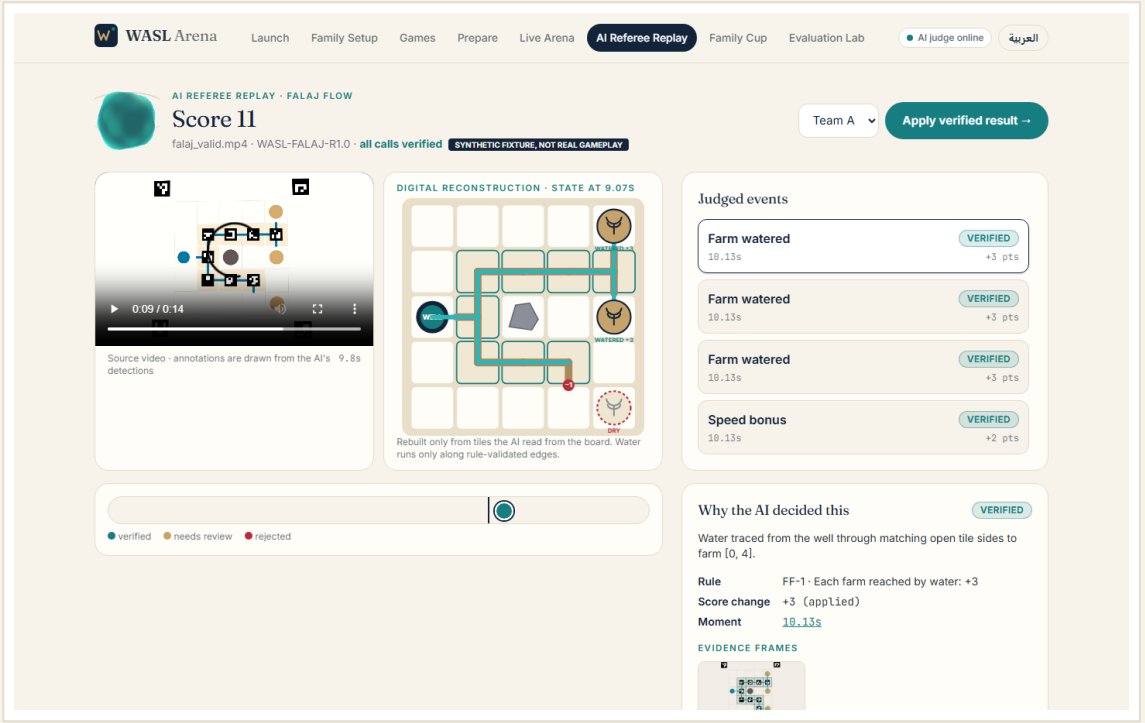
After

- Rule FB-1: ball observed entering the goal zone after being seen outside: **+1** · Ball lost near the goal → **POSSIBLE_GOAL**, 0 points until the host decides

GAME 2 · TEAM

Falaj Flow: overview

- Inspired by the UAE **falaj** irrigation system
- Printed 5×5 board: **well**, three **farms**, a **rock**
- Teams place channel tiles for **90 seconds**
- Farm **+3** · leak **–1** · speed bonus up to **+2**
- Tie: shortest valid network, then time



Split screen: physical board and digital twin

Falaj Flow: graph intelligence

1 Calibrate

4 corner markers → board homography

2 Read tiles

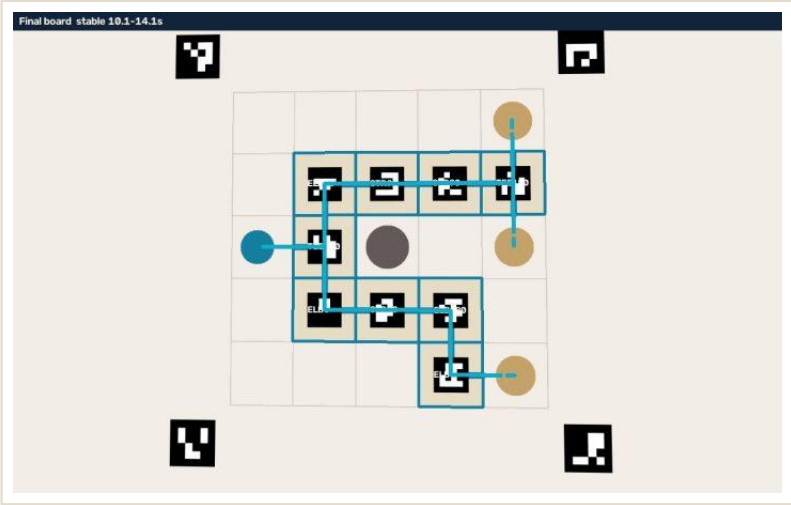
marker id = type, rotation = 90° steps

3 Stabilise

state must hold 1 s: hands ignored

4 Trace water

BFS over matching open sides; leaks found



Board read from video, water path in teal

- Exact tile types and rotations recovered in testing
- Leaks: edge, rock, gap, closed side

RULES

Falaj Flow: scoring

Farm watered	+3
Leak	–1
All farms ≤ 45 s	+2
All farms ≤ 70 s	+1
Tie-break	shortest network → time

Tested outcomes

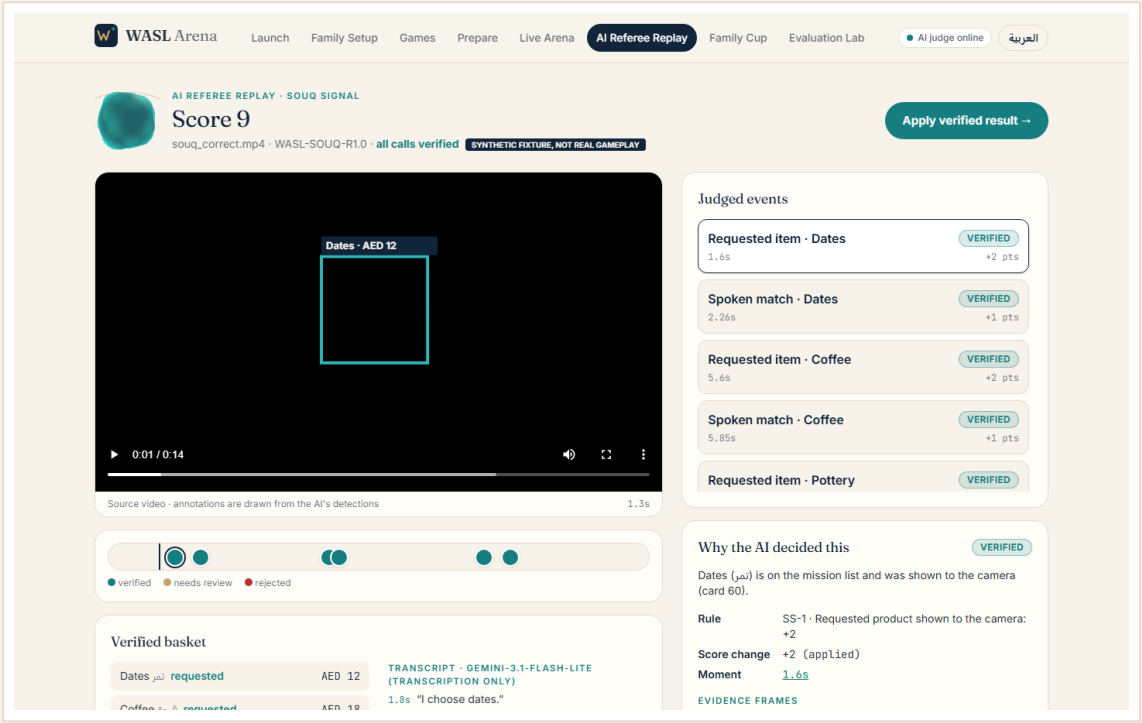
falaj_valid: 11 pts
falaj_partial: 5 pts
falaj_rock: 10 pts

Complete route 11 · partial network 5 · blocked by rock 10 (synthetic fixtures, matching ground truth).

GAME 3 · INDIVIDUAL

Souq Signal: overview

- Shopping **mission** + **budget** in AED
- Pick product cards: dates, coffee, spices, pottery, textile, oud…
- **Show** each card to the camera and **say** its name
- Item **+2** · spoken match **+1** · over budget **–2**
- Duplicate = no points · Tie: lower spend, then time



Verified basket and transcript

VISION + SPEECH

Souq Signal: multimodal AI

See

Card markers decoded per frame → presentation intervals

Hear

Arabic/English speech → timed transcript

Agree?

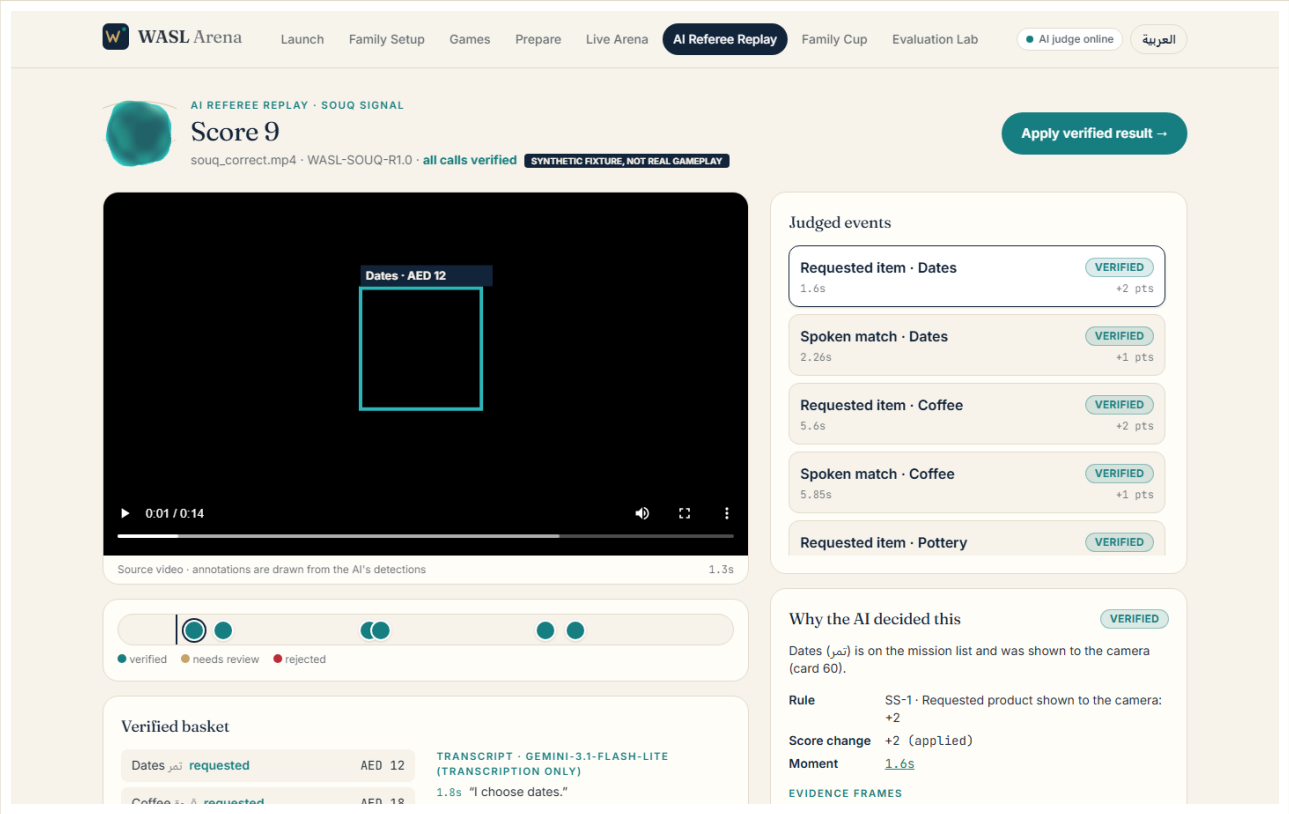
Said ≈ shown within ±5 s, else host review

Score

Deterministic rules + budget maths

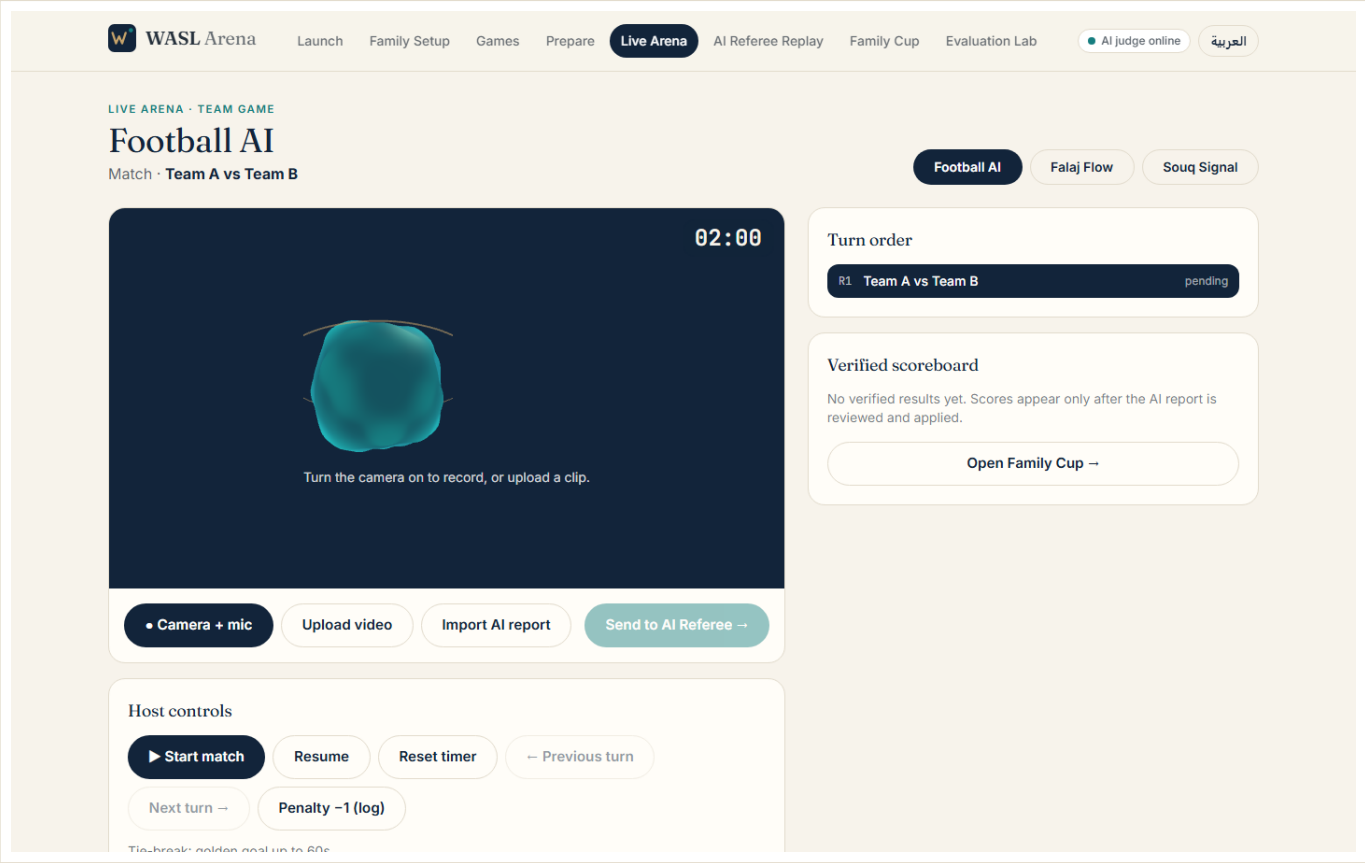
- Transcription model **never decides points**
- Silent audio is rejected by an energy gate (no hallucinated speech)
- No transcript → spoken-match points stay **NEEDS_REVIEW**

Souq Signal: transaction verification



- Distinct card prices are summed
- Over budget → **−2** (once)
- Duplicate cards earn **0** and are not paid twice
- Audio ≠ video → **host review**
- Test: correct basket = 9 pts, spend AED 55 of 60

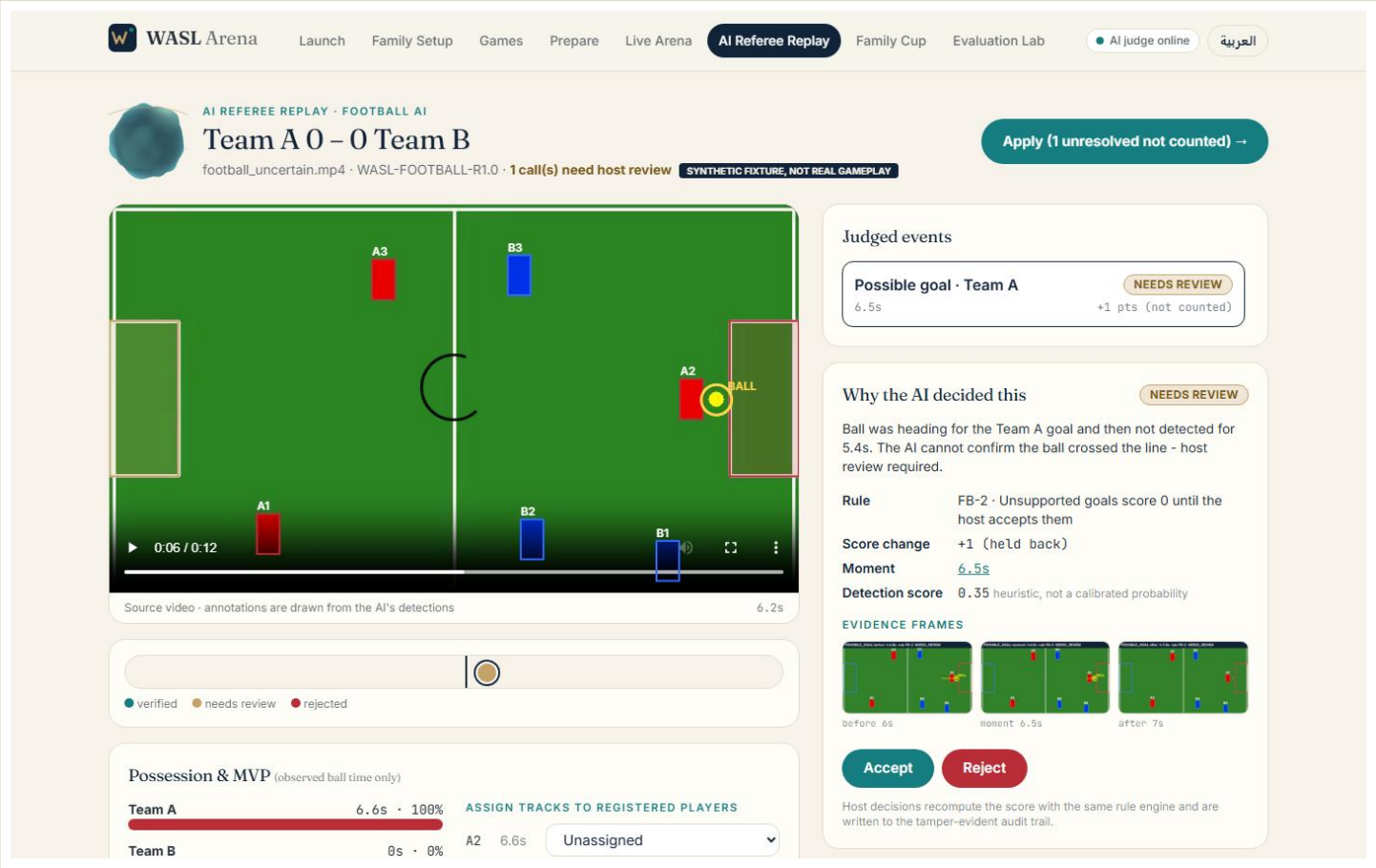
Family host console



- Camera + mic recording or **upload**
- Round **timer**, turn order, **skip/next**
- Import a sealed **AI report**
- Verified **scoreboard**
- Host penalty with audit log

EXPLAIN EVERY CALL

AI Referee Replay



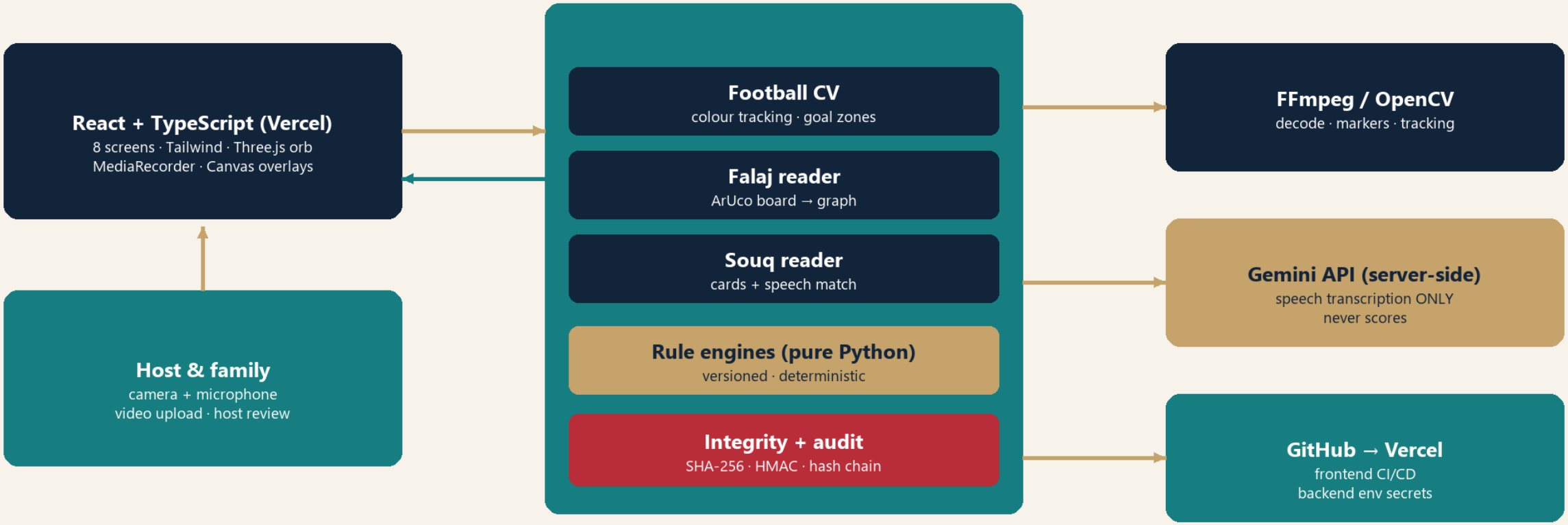
A call that needs review: ball lost near the goal

- Event timeline on the video
- **Why the AI decided this**
- Evidence frames + audio transcript
- **Accept / Reject** → score recomputed
- Audit trail entry for every decision

SYSTEM

Technical architecture

System architecture



METHODS

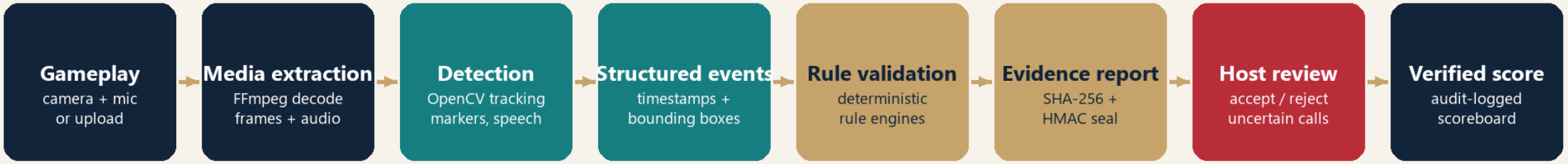
AI models and algorithms

Football	HSV segmentation, contours, nearest-neighbour tracking, goal-zone logic
Falaj Flow	ArUco markers, homography, rotation estimation, BFS on tile graph
Souq Signal	ArUco card sightings, speech transcription, alias matching, budget maths
Scoring	Versioned deterministic rule engines (FB / FF / SS)
Integrity	SHA-256, HMAC-SHA256, hash-chained audit log

TRUST

Evidence integrity

AI judging pipeline: every stage leaves evidence



- Source file **SHA-256** and rule-version id on every report
- Report sealed with **SHA-256 + HMAC**; import of an edited report is **rejected** (tested)
- Host actions in a **hash-chained audit log**; tampering detected (tested)
- Stand-alone Python judge produces the same sealed report

RESPONSIBLE AI

Fairness and privacy

Fairness

Rules are written and versioned
Uncertain = NEEDS_REVIEW, never guessed
Unobserved ball time is 'unknown'
Every call can be accepted / rejected
Arabic + English, team + solo

Privacy & security

No face identification or biometrics
Credentials only in server environment
Audio sent only for transcription
Consent from everyone filmed
Reports signed against tampering

HOW WE TEST

Evaluation methodology

Rule engines

Unit tests per game: valid, invalid, edge, tie-break

Pipelines

Synthetic fixtures with known ground truth → video → events → score

API & integrity

Upload → job → report → host review → tamper rejection

Honest scope: fixtures are rendered by code, not real family gameplay. They prove the pipelines work; they do not measure real-world accuracy. Real clips are the next step.

VERIFIED BY RUNNING

Actual test results

46/46

automated tests passed

9/9

synthetic fixtures match ground truth

0

real family clips analysed yet

football_goal ✓

goals_A=1, goals_B=0, verified_goal_events=1

football_nogoal ✓

goals_A=0, goals_B=0, verified_goal_events=0

football_uncertain ✓

goals_A=0, goals_B=0, verified_goal_events=0

falaj_valid ✓

destinations=3, leaks=0, time_bonus=2

falaj_partial ✓

destinations=2, leaks=1, time_bonus=0

falaj_rock ✓

destinations=3, leaks=1, time_bonus=2

souq_correct ✓

total=9, spend=55, mismatch=False

souq_mismatch ✓

total=2, spend=12, mismatch=True

souq_budget_dup ✓

total=7, spend=55, mismatch=False

DELIVERY

GitHub and Vercel deployment

GitHub

MayMourad/wasl-arena (private): frontend, backend, tests, docs, AI-usage records

Vercel · frontend

frontend-three-fawn-47.vercel.app · VITE_API_URL points at the judge

Render · AI judge

wasl-arena.onrender.com · FastAPI + FFmpeg + OpenCV · secrets as env vars

- **Verified live:** all routes load, badge shows “AI judge online”, a Falaj clip judged on the live backend = 11 pts, seal valid
- CORS locked to the Vercel origin · no secrets in the repository (.env git-ignored)
- Free tier: first request after idle can take ~30 s

NEXT

Limitations and future development

Limitations

No real family footage evaluated yet
Controlled setup: fixed camera, bright ball
Colour segmentation is lighting-sensitive
Printed markers can be hidden or damaged
Speech timing ±~1 s; scores are heuristics

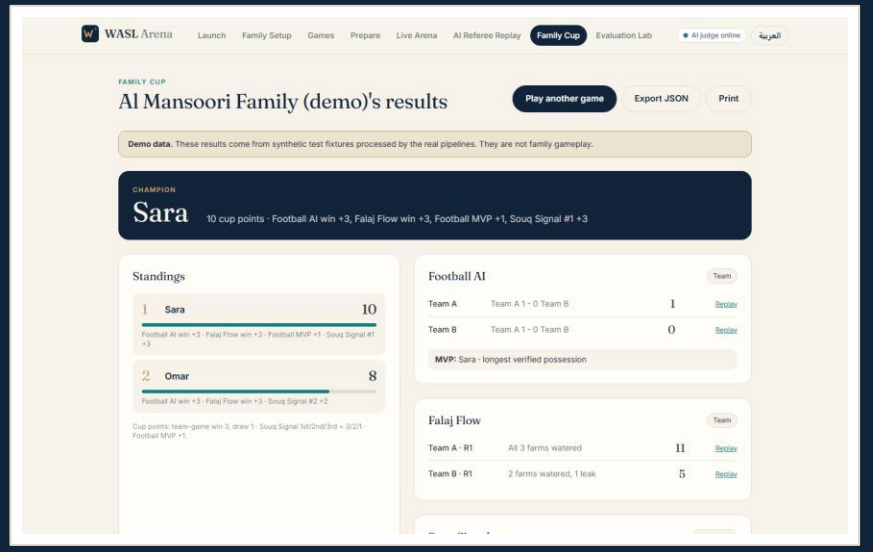
Future development

Evaluate real footage for each game
Pretrained detectors + jersey numbers
Whistle / 'goal!' audio cues
Host-defined games from the rule template
More heritage games and languages

WASL Arena

A fair referee shows its evidence, admits doubt, and lets people decide.

- Three games · team and individual
- Real video/audio judging, explainable results
- Interactive, inclusive, culturally meaningful, transparently fair



Team 204 · Code AYUCK · Cycle 3 (grades 9-12)
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